

Highway Congestion Reduction through RL-Based Eulerian Headway Control

Control density through headway, not speed.

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Main contributions

CAV-based highway congestion reduction framework that:

- 1** Controls traffic density through headway
- 2** Is designed for practical deployment using existing technologies

Background

Flow (2018) – Cathy Wu, Alex Bayen et al.



Distributed control (2021)– Jiaxun Cui, Peter Stone et al.



CIRCLES MegaVanderTest experiment (2022)



Collaborators

Berkeley
UNIVERSITY OF CALIFORNIA

BAIR
BERKELEY AUTONOMOUS INTELLIGENCE RESEARCH

Berkeley DeepDrive

RUTGERS
UNIVERSITY OF NEW JERSEY

Temple University

THE UNIVERSITY OF ARIZONA

VANDERBILT UNIVERSITY

Triangle Research Center

Université Gustave Eiffel

TN
Department of Transportation

TOYOTA

NISSAN
MOTOR CORPORATION

gm
RESEARCH & DEVELOPMENT

amazon
web services

SIEMENS

CIRCLES: coordinating CAVs to smooth real-world traffic

CIRCLES

Using deep reinforcement learning and CAVs to improve traffic flow and reduce energy consumption.



CIRCLES

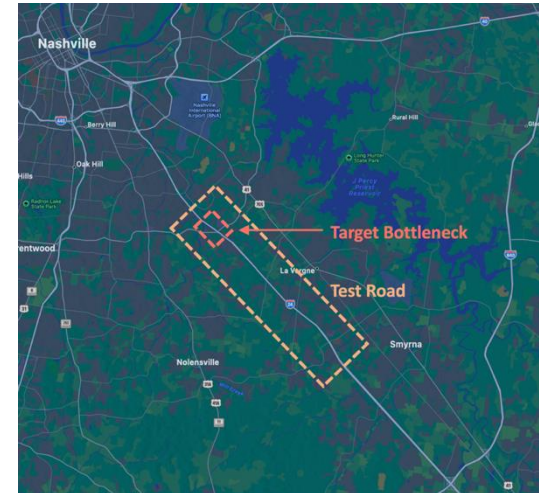


MegaVanderTest

- The final open-road test of the CIRCLES project – sending 100 AVs onto I-24 during the morning peak
- One of the largest coordinated open-road traffic-smoothing experiments conducted to date
- 8% reduction in avg traffic density, 52% reduction during congestion formation

[Lee et al., IEEE Control Systems 2025]

[Wang et al. IEEE Control Systems 2025]

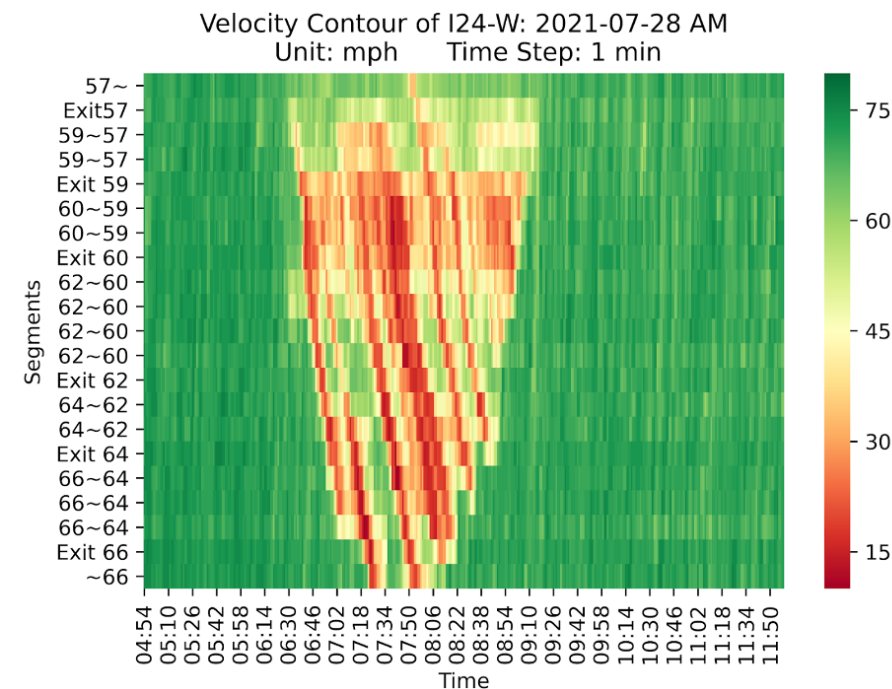


A single I-24 camera pole

This paper: congestion reduction in realistic multilane highway bottlenecks



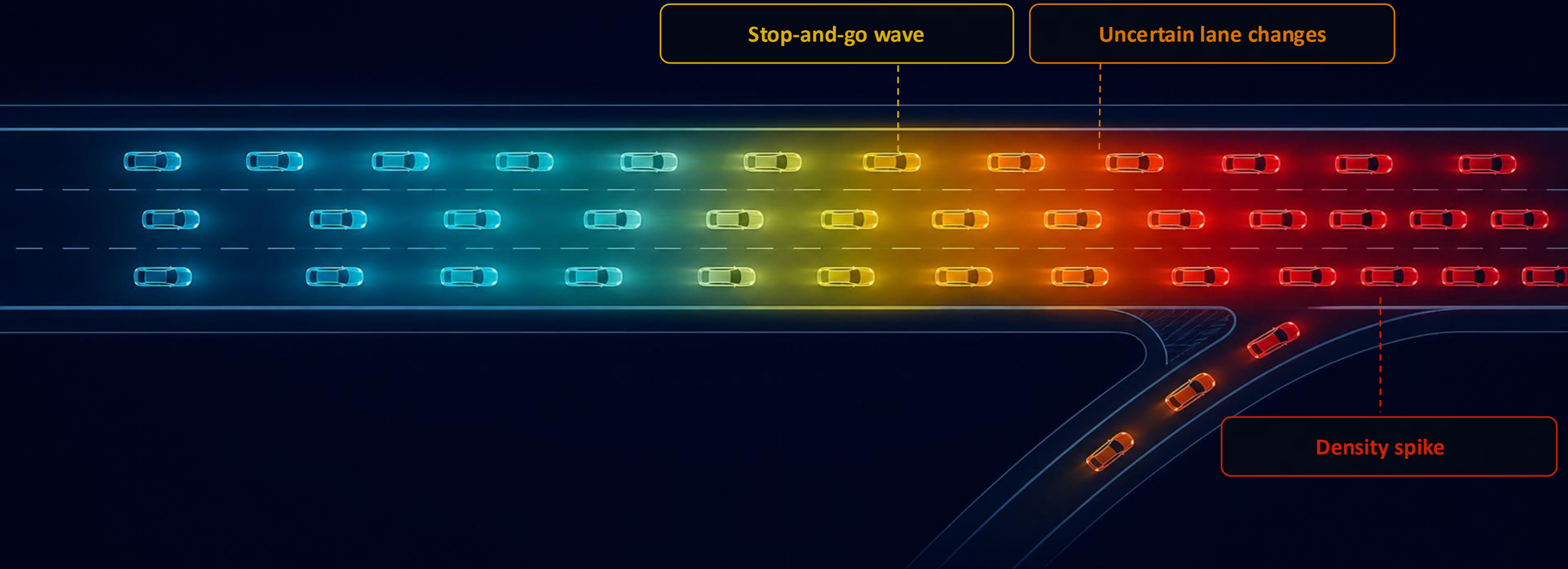
MVT Bottleneck: a high-demand weaving area between Exit 60 and Exit 59 on I-24 westbound.



INRIX Heatmap: average-speed profile on the test road segment, based on data provided by INRIX, Inc.

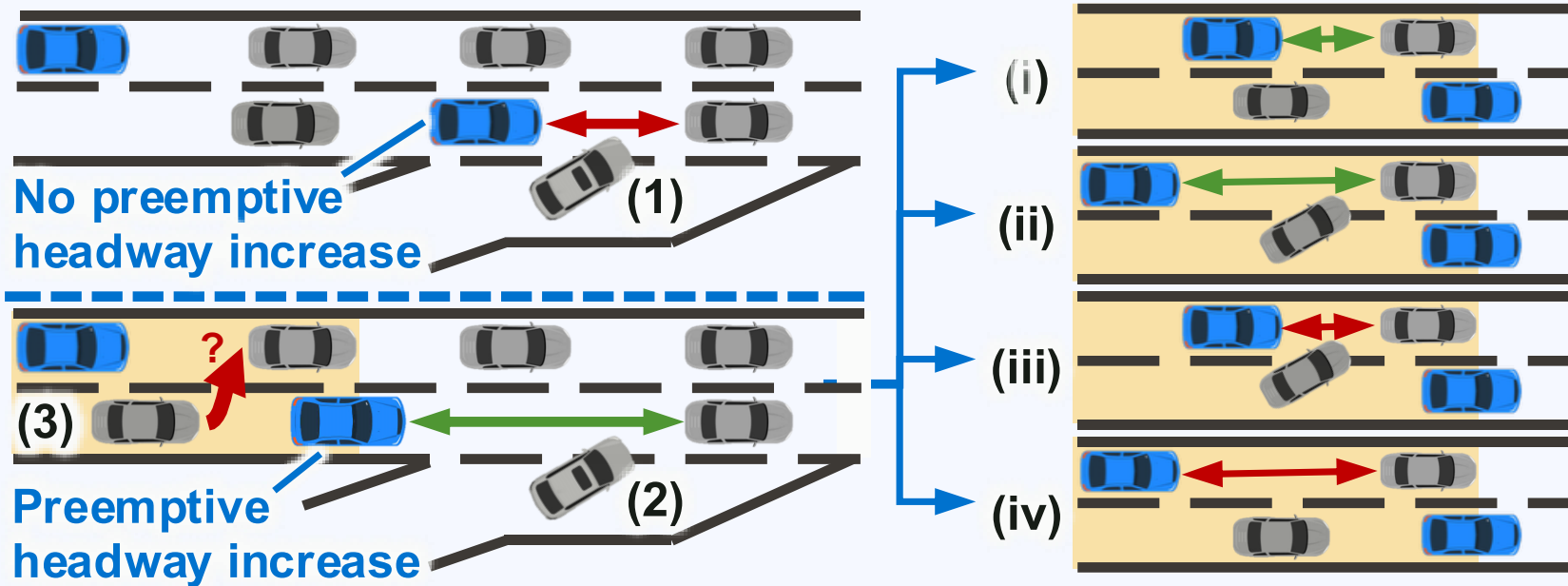
Target bottleneck: Exit 59 on I-24 West, Tennessee

A local merge can trigger a network-wide breakdown



- Headway increase helps if the location and time of lane-changes are predictable
- However, in multilane bottlenecks the location and timing of lane-changes are **highly uncertain**

Vehicle-level decisions require access to other drivers' hidden intentions



Increase headway is useful **now** if and only if a lane-change **will** occur

Robust action would require predicting counterfactual driver behavior → Motivates density regulation

Claim: connectivity & automation can be better utilized

Existing systems



Eulerian VSL¹

Roadside signs · aggregate control

- ✗ Low responsiveness — signs update only a few times per hour
- ✗ Inconsistent human compliance
- ✓ Doesn't depend on intention-prediction

Recently proposed systems



Lagrangian vehicle² control

Direct vehicle actuation

- ✓ High responsiveness
- ✓ High compliance
- ✗ Depends on latent lane-change intentions in multi-lane traffic

vs



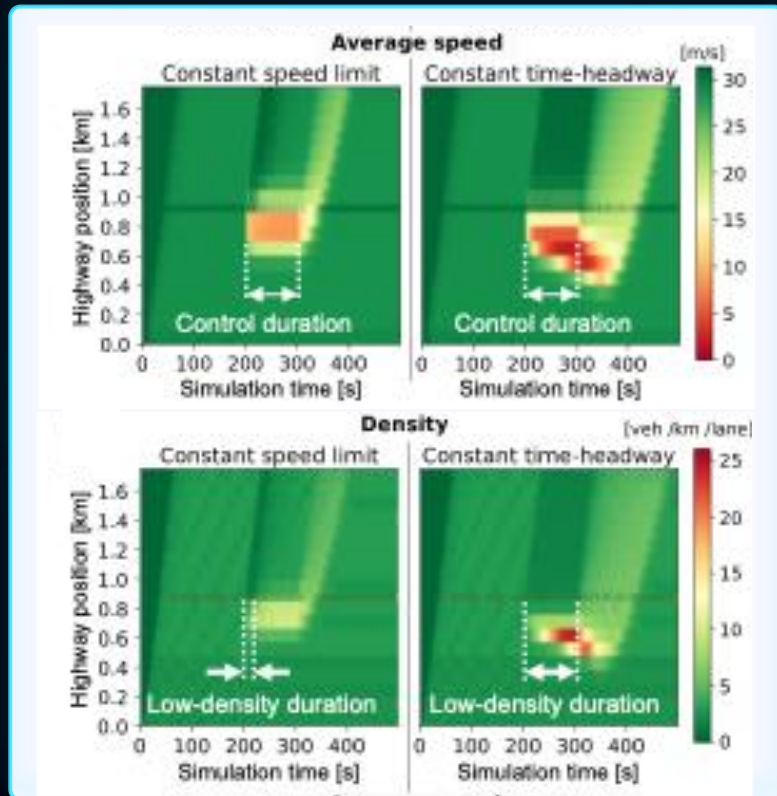
Design target: **responsive + compliant + intention-agnostic**

¹ [Papageorgiou 2023, Hegyi 2005, Han 2017]

² [Vinitsky et al. 2023, Cui et al. 2021, Zhang et al. 2023]

Density regulation dodges intention-prediction. Headway directly regulates density.

Constant speed limit vs. constant time-headway



◀ Average speed: green is fast, yellow/red is slow.

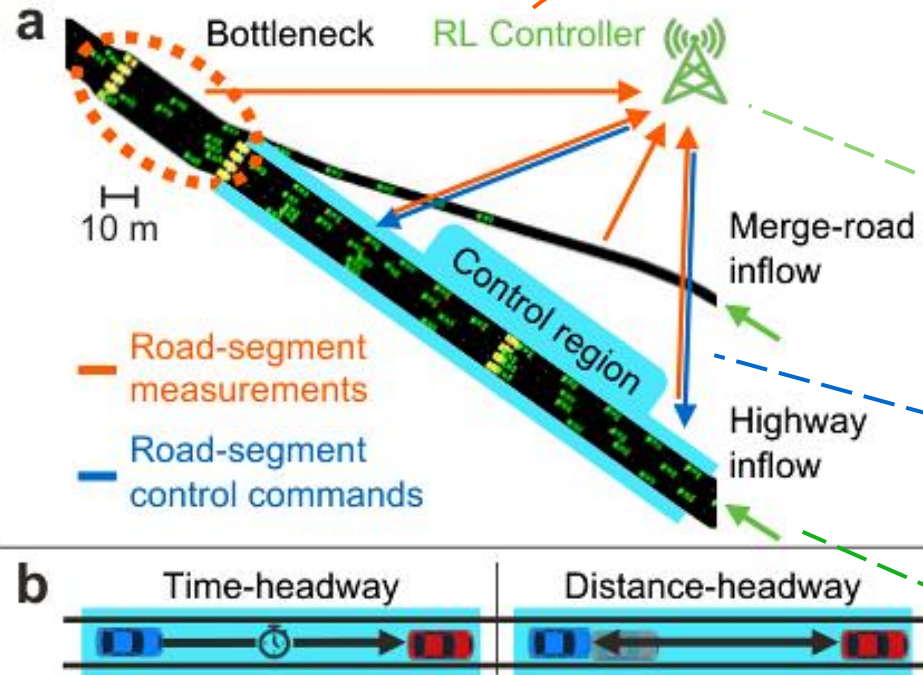
White dotted regions: Duration for which the control actively shapes traffic

◀ Density: green is sparse, yellow/red is dense.

White dotted regions: Duration of the downstream low-density region

- Time-headway commands set vehicle spacing directly, sustaining a low-density region
- In contrast, achieving the same effect through speed commands requires carefully timed dynamic control

Proposed System



1 Sense segment speed + density

2 Infer RL-based headway actions

3 Broadcast actions to Connected ACC veh.

4 ACC adjusts headway within safeguards

Safety is assumed to be enforced by production ACC; commands are constrained to the allowable range

Practical by design: designed to run on the ACC you already have

1

Adaptive Cruise Control (ACC) integration

Hooks directly into ACC, adjusting a single headway parameter

2

Existing traffic perception & communication

I-24 MOTION-like system, low-bandwidth (cellular) communication

3

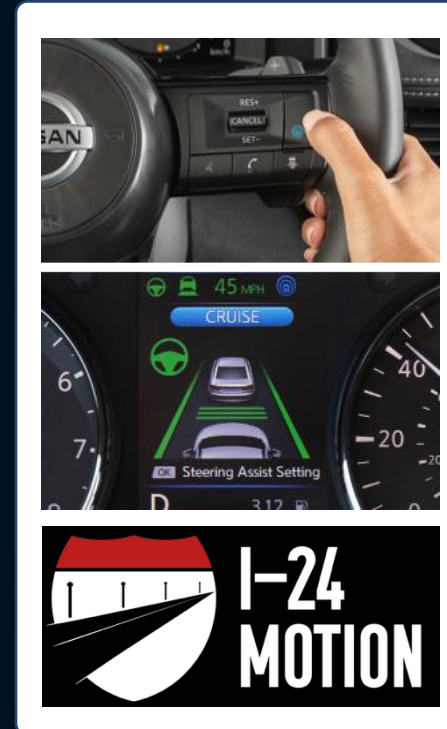
Practical & scalable deployment

One headway command per segment + local independent deployment

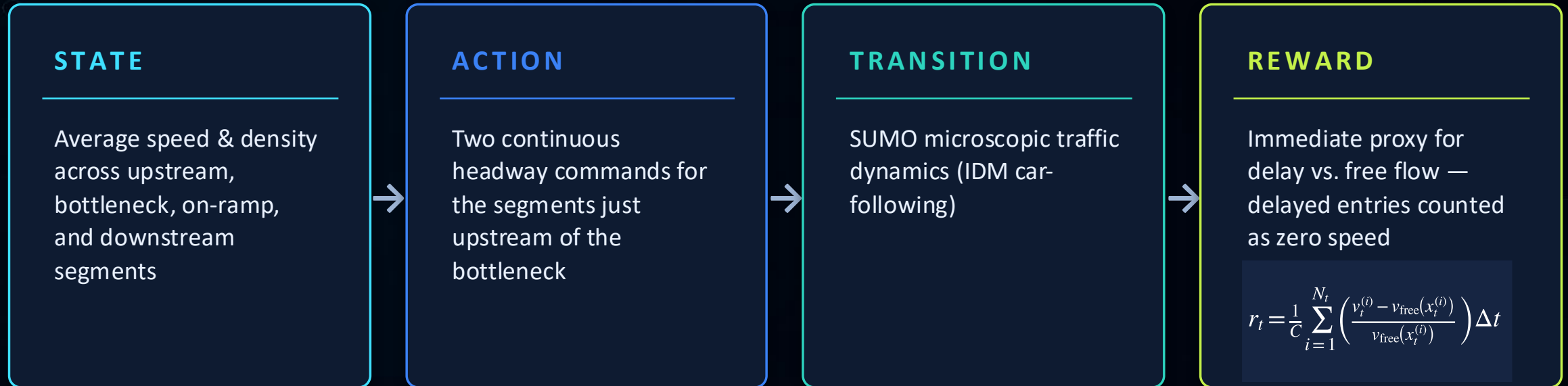
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Leveraging built-in ACC safeguards

Headway commands restricted to ACC-allowed ranges



RL learns headway commands from segment-level traffic fields



TRAINING PPO/SAC via RLlib in SUMO

Evaluation setup

~1,000

vehicles simulated

500 s

episode length

30

random seeds

95%

confidence intervals

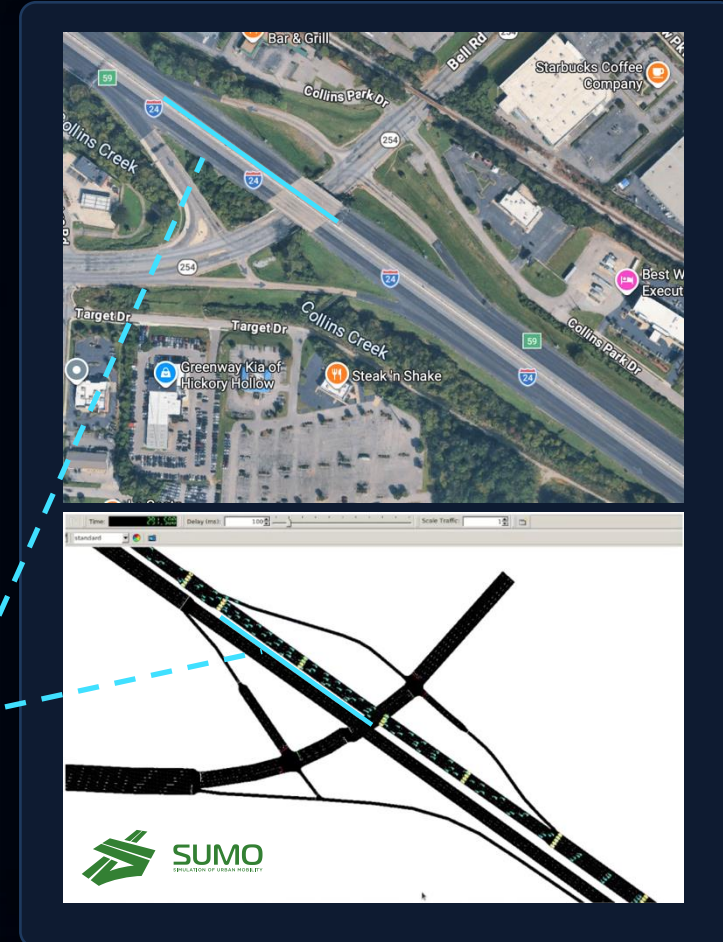
Merge inflow **7 / 14 / 30 veh/min**

ACC penetration **20% / 60% / 100%**

Baselines:

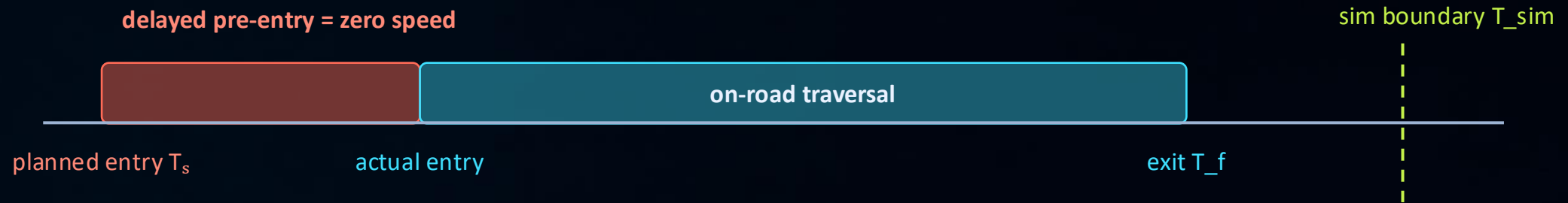
1. Human-driven
2. Traditional VSL
3. Dynamic RL-based VSL

Two controlled segments



*Simulated 2 km I-24 merge:
four lanes, one on-ramp,
two controlled upstream segments*

Boundary-aware metric penalizes delayed entry



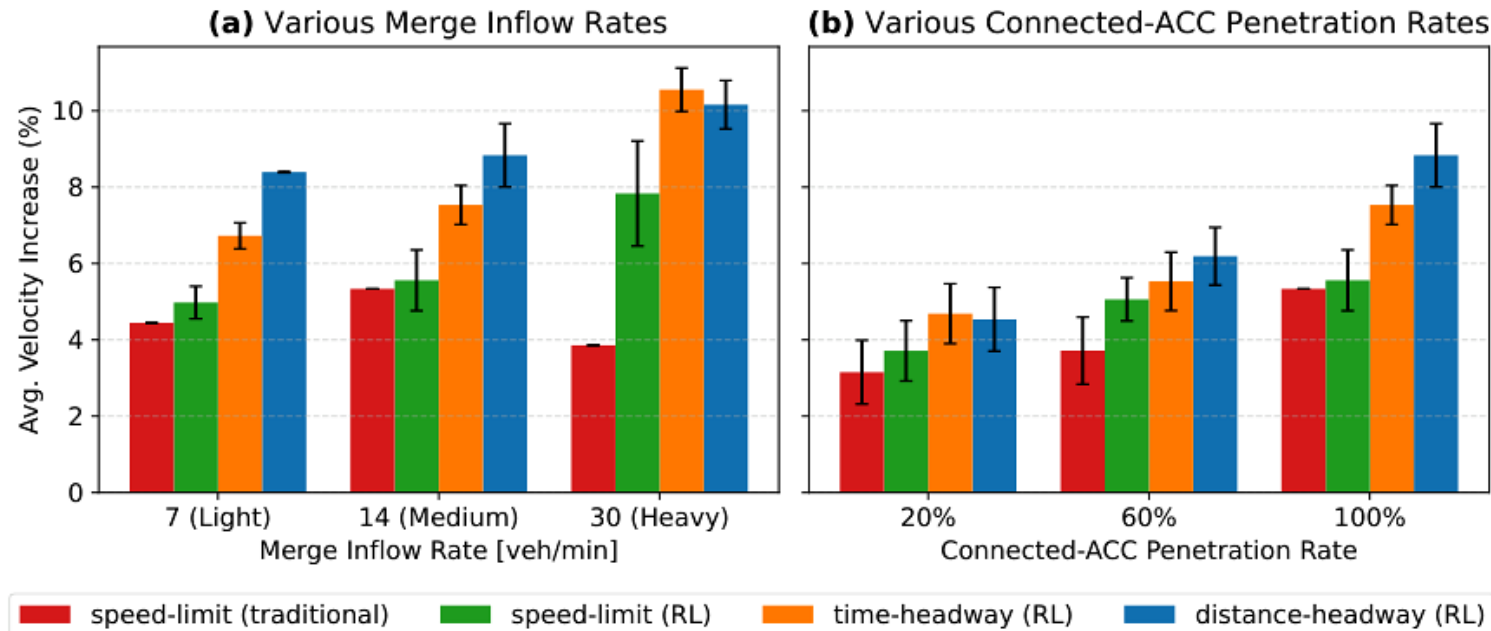
A previously recognized limitation [Cui et al, AAMAS 2021] – finite-road simulations can be gamed:
Naive average speed metric ignores upstream waiting before entering the simulation

Our novel boundary-aware metric counts delayed time as zero speed:

$$\bar{v}^{(i)} = \frac{L^{(i)}}{\min(T_f^{(i)}, T_{sim}) - T_s^{(i)}}$$

→ A controller can no longer score well merely by delaying vehicles entry to the simulated road

Headway control consistently outperforms speed-based baselines



+10.6%

vs. human traffic

+6.7%

vs. traditional VSL

+3.4%

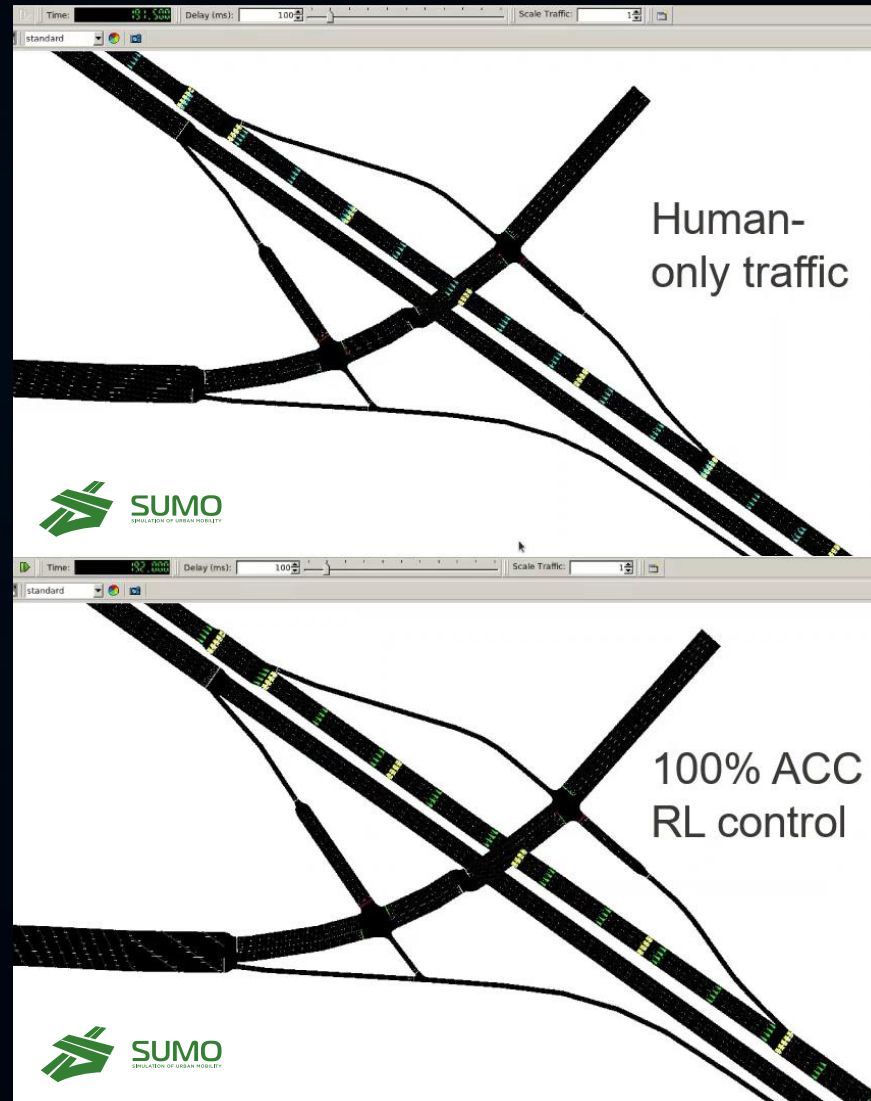
vs. RL-based VSL

Maximum gains at 100% connected-ACC penetration

- Headway-based control consistently outperforms speed-based baselines
- More compliant ACC vehicles → stronger density regulation → larger speed gains

Headway control consistently outperforms speed-based baselines

- Example video comparison:



Summary

Sense aggregate traffic



Command headway



Reduce bottleneck congestion

Contributions

- Responsive Eulerian **headway control**
- No dependence on counter-factual lane-change intentions
- Uses existing connected-ACC capabilities
- **Deployment-facing design**

Future work

- Calibrate simulation with real-world data
- Validate on-road
- Combine with ramp metering & dynamic lane assignment

Control the traffic density. Let vehicles self-organize.

Questions?

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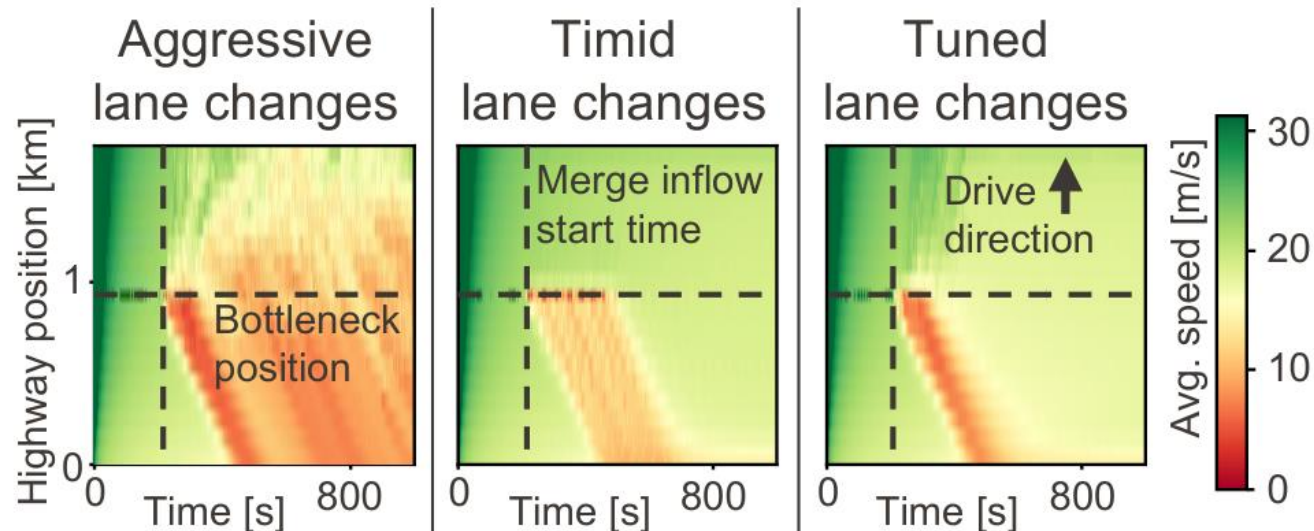
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<https://coopcruise.github.io/>



Lane-change calibration avoids unrealistic congestion extremes

BACKUP



- 1 Horizontal axis: simulation time
- 2 Vertical axis: highway position
- 3 Color: average speed — green is faster; yellow/red is slower
- 4 Dashed markers: merge inflow start and bottleneck position

Aggressive lane changes

Frequent disruptive interactions create persistent congestion around the bottleneck.

Timid lane changes

Insufficient redistribution produces prolonged right-lane slowdown and effectively single-lane behavior.

Tuned lane changes

Balanced lane use produces a localized disturbance followed by flow recovery.

Calibration is essential: evaluate the controller against plausible traffic dynamics — not pathological lane-changing extremes.

Lane-change calibration avoids unrealistic congestion extremes

BACKUP

